

EARTHPRESS · REQUIREMENTS

EarthPress requirements

Open Hardware Portfolio · amishchadha.com

DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/earthpress		
Source	docs/03-requirements.md @ ab4786d		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept estimates
0.1	2026-09-25	Amish Chadha	Initial scaffold

EarthPress requirements

These are first-pass requirements for the concept. Targets are proposals for review, not yet validated with users, and will be checked by calculation at TRL 3 and revised after co-design sessions (see EPR-PRB-001). The status column gives the concept estimate from EPR-PRC-001; every value there is an estimate.

The **reference block** is 290 x 140 x 90 mm (proposed, awaiting Amish), about 3.65 L, from a sandy clay loam stabilized with 5 % cement by dry mass.

ID	Requirement	Target	Verification (TRL 3 or later)	Concept status
R1	Block dimensions	290 x 140 mm $\pm$ 2 mm on plan; height 90 mm $\pm$ 3 mm, checked with the block gauge	Mold tolerance analysis; later gauge checks on a production run	Met by design, unverified
R2	Compaction pressure	2.0 MPa or more on the block face at the end of the stroke	Linkage force calculation	At risk: about 81 kN needed. Force ratio is reachable (lever about 19:1 times toggle about 9:1 within about 3° of straight), but the compaction work, about 0.6 to 1.2 kJ, exceeds the about 0.65 kJ one operator delivers over the modeled 50° lever arc
R3	Block strength with suitable soil	28-day dry compressive strength 4 MPa or more and wet strength 2 MPa or more (Auroville class B); in all cases 2.1 MPa (300 psi) or more, as in 14.7.4 NMAC	Literature review at TRL 3; block tests at TRL 4	Not verifiable at TRL 2. Depends on soil, mix and curing as much as on the press
R4	Output	300 or more blocks in an 8 h day with a crew of four, including mixing	Cycle time estimate; later timed trials	Met on paper (about 96 s per block allowed), unverified
R5	Operator force	Peak pull of 500 N or less at the lever grip; grip between 0.9 and 1.6 m above ground through the stroke	Force calculation; ergonomic check of the lever arc	At risk: in the modeled layout the grip starts about 1.36 m high and ends about 0.3 m above ground, below the 0.9 m target
R6	Garage-buildable	Built with a stick welder, angle grinder, drill press and hand tools; no turned or milled parts except purchased pins and bushings	Design review of every part	Met by design
R7	Movable	Complete press 140 kg or less; no single piece over 50 kg after removing pins and bolts; fits a 1.5 m pickup bed	Mass estimate from the model	Not met: press about 123 kg, but the frame alone is about 61 kg
R8	Durability	50,000 blocks before mold liners or bushings need replacing; all pins in replaceable bushings; grease points on every pivot	Wear estimate; later endurance log	Not assessed at TRL 2
R9	Soil test kit	Field tests (jar sedimentation, shrinkage box, ribbon and drop tests) give a go or no-go and a starting cement content, with a first screen within 1 h and a full result within 24 h	Written procedure; later comparison with lab grading on partner soils	Met by design, unverified
R10	Low stabilizer use			Plausible from Auroville practice, unverified

ID	Requirement	Target	Verification (TRL 3 or later)	Concept status
		Blocks reach R3 with 8 % cement by dry mass or less; 5 % as the design target	Literature at TRL 3; block tests at TRL 4	
R11	Embodied carbon	Cradle-to-gate CO2 of the walling material 25 % or less of a fired-brick wall of equal volume	Carbon estimate from published factors	Met on paper: about 8 % using Auroville figures
R12	Safe operation	Lever cannot fall freely when released under load; lid latch holds against at least 1.5 times the peak block force; pinch points between lever and frame guarded or with 100 mm or more clearance	Design review; later proof-load test	Addressed in concept; not yet dimensioned
R13	Affordable	Press and test kit \$450 or less in parts, excluding labor and cement	Priced BOM (bom/bom.csv)	Met: about \$403 (indicative)
R14	Open and documented	CERN-OHL-S drawings, BOM and build notes sufficient for a welder to build without contacting the author	Review by an outside fabricator at TRL 3	Not yet assessed

■ Requirements not met or at risk

- **R7 not met:** the frame with its 20 mm base plate is about 61 kg against the 50 kg single-piece limit. Options are a thinner base plate or a bolted two-part frame.
- **R2 at risk:** the force ratio can reach 2 MPa, but the work needed may exceed what one person delivers over a 50° lever arc. Raising the lever pivot to allow about a 100° arc, or two people on the lever, would cover it. Reaching 2 MPa also depends on the toggle ending within about 3° of straight, which makes the final pressure sensitive to fill mass.
- **R5 at risk:** the lever grip ends about 0.3 m above ground, which forces a deep stoop at the end of the stroke, where force is highest. The same raised pivot would help.
- **R3, R8 and R10 cannot be shown at TRL 2.**

■ Assumptions

- Compaction pressure of 2 to 4 MPa and about 5 % cement follow Auroville Earth Institute practice ([Auroville Earth Institute](#)).
- Dry block density about 1,900 kg/m³ (estimate), giving about 6.9 kg per dry block.
- One operator can pull 500 N at the grip for short strokes, using body weight; this is an assumption to be checked with users.
- A crew of four covers soil sieving, mixing, pressing and stacking.